

MATH3881/5881: Statistical Machine Learning Theory

Week 0: Basics of Calculus

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Key topics

- Vectors and Functions
- Derivatives: Directional Derivative, Gradient, Jacobian, and Hessian
- The Multivariable Chain Rule
- Jacobian-Vector Product and Vector-Jacobian Product
- Taylor's Theorem

Reference:

- Appendix A of *Mathematical Engineering of Deep Learning* by Lique, Moka, and Nazarathy: <https://deeplearningmath.org/>

0.1.1 Vectors and Functions in $\mathbb{R}^n$

Notation

- $\mathbb{R}$: The set of all real numbers.
- $\mathbb{R}^n$: The n -dimensional real coordinate space, where each element is a vector represented as a column:

$$u = (u_1, \dots, u_n) = [u_1 \ \cdots \ u_n]^\top = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}.$$

Euclidean Norm and Inner Product

- The **Euclidean norm** (or L_2 norm) of $u \in \mathbb{R}^n$ is $\|u\|_2 = \sqrt{u^\top u} = \left(\sum_{i=1}^n u_i^2 \right)^{1/2}$.
- The **inner product** of two vectors $u, v \in \mathbb{R}^n$ is the scalar $u^\top v = \sum_{i=1}^n u_i v_i$.
- The **cosine of the angle** between u and v is:

$$\cos \theta = \frac{u^\top v}{\|u\|_2 \|v\|_2}. \quad (0.1)$$

L_p Norm

- For $p \geq 1$, the L_p norm of $u \in \mathbb{R}^n$ is

$$\|u\|_p = \left(\sum_{i=1}^n |u_i|^p \right)^{1/p}.$$

- The default norm $\|u\|$ (without subscript) refers to the L_2 norm.

Key Inequalities

- **Cauchy-Schwarz inequality:** For any $u, v \in \mathbb{R}^n$,

$$|u^\top v| \leq \|u\| \|v\|, \quad (0.2)$$

with equality **if and only if** u and v are linearly dependent (i.e., $u = cv$ for some $c \in \mathbb{R}$).

- **Triangle inequality:** For any $u, v \in \mathbb{R}^n$, we have $\|u + v\| \leq \|u\| + \|v\|$.

Exercise 0.1.1.1

By expanding $\|u + v\|^2$ and using the Cauchy-Schwarz inequality, establish the triangle inequality.

Euclidean Distance

- The distance between $u, v \in \mathbb{R}^n$ is $\|u - v\| = \left(\sum_{i=1}^n (u_i - v_i)^2 \right)^{1/2}$.

Convergence in $\mathbb{R}^n$

- A sequence $\{u^{(k)}\}$ in $\mathbb{R}^n$ **converges** to a vector $u \in \mathbb{R}^n$ (denoted $u^{(k)} \rightarrow u$) if $\lim_{k \rightarrow \infty} \|u^{(k)} - u\| = 0$.
- That is, for every $\varepsilon > 0$, there exists a positive integer N_0 such that $\|u^{(k)} - u\| < \varepsilon$ for all $k \geq N_0$.

Exercise 0.1.1.2

Consider the sequence of vectors $\{u^{(k)}\}$ in $\mathbb{R}^2$ defined by

$$u^{(k)} = \left[\frac{1}{k}, 2 + \frac{3}{k} \right]^\top \quad \text{for } k = 1, 2, 3, \dots$$

Then,

- Identify the proposed limit vector $u \in \mathbb{R}^2$ for this sequence. Justify your answer using component-wise limits.
- Compute the Euclidean distance $\|u^{(k)} - u\|$ and show that

$$\lim_{k \rightarrow \infty} \|u^{(k)} - u\| = 0.$$

- Generalize your reasoning: In $\mathbb{R}^n$, if a sequence $\{u^{(k)}\}$ satisfies $\lim_{k \rightarrow \infty} u_i^{(k)} = u_i$ for every component $i = 1, \dots, n$, prove that $u^{(k)} \rightarrow u$ in the Euclidean norm.

Continuity of Functions

- **Scalar functions** ($f : \mathbb{R}^n \rightarrow \mathbb{R}$):

- f is **continuous at** u if for **every** sequence $u^{(k)} \rightarrow u$,

$$\lim_{k \rightarrow \infty} f(u^{(k)}) = f(u).$$

- Or, equivalently, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(u) - f(v)| < \varepsilon \quad \text{whenever} \quad \|u - v\| < \delta.$$

- **Vector-valued functions** ($f : \mathbb{R}^n \rightarrow \mathbb{R}^m$):

- Expressed as

$$f(u) = [f_1(u) \ \cdots \ f_m(u)]^\top, \quad (0.3)$$

where each component $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a scalar function.

- f is **continuous at u** if **each** f_i is continuous at u .
- f is **continuous on a set $\mathcal{U} \subseteq \mathbb{R}^n$** if it is continuous at every $u \in \mathcal{U}$.

0.1.2 Derivatives

Partial Derivatives and Gradient

- For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **partial derivative** with respect to u_i is

$$\frac{\partial f(u)}{\partial u_i} = \lim_{h \rightarrow 0} \frac{f(u_1, \dots, u_i + h, \dots, u_n) - f(u)}{h}. \quad (0.4)$$

- The **gradient** of f at u aggregates all partial derivatives as

$$\nabla f(u) = \left[\frac{\partial f(u)}{\partial u_1}, \dots, \frac{\partial f(u)}{\partial u_n} \right]^\top. \quad (0.5)$$

- For matrix inputs $U \in \mathbb{R}^{n \times m}$, the gradient is structured as

$$\frac{\partial f(U)}{\partial U} = \begin{bmatrix} \frac{\partial f(U)}{\partial u_{1,1}} & \cdots & \frac{\partial f(U)}{\partial u_{1,m}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f(U)}{\partial u_{n,1}} & \cdots & \frac{\partial f(U)}{\partial u_{n,m}} \end{bmatrix}. \quad (0.6)$$

Directional Derivatives

- The **directional derivative** of f at u in direction $v \in \mathbb{R}^n$ is

$$\nabla_v f(u) = \lim_{h \rightarrow 0} \frac{f(u + hv) - f(u)}{h}.$$

- Key relationships:
 - Partial derivatives are directional derivatives along coordinate axes: $\nabla_{e_i} f(u) = \frac{\partial f(u)}{\partial u_i}$.
 - For differentiable f , we have $\nabla_v f(u) = v^\top \nabla f(u)$.
 - Among all directions $v \in \mathbb{R}^n$ with unit norm ($\|v\| = 1$), the directional derivative is maximized when $v \propto \nabla f(u)$, by the Cauchy-Schwarz inequality.

Exercise 0.1.2.1

Prove the three key relationships stated above.

Jacobians

- For $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with $f(u) = [f_1(u), \dots, f_m(u)]^\top$, the **Jacobian** is

$$J_f(u) = \begin{bmatrix} \frac{\partial f_1(u)}{\partial u_1} & \dots & \frac{\partial f_1(u)}{\partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(u)}{\partial u_1} & \dots & \frac{\partial f_m(u)}{\partial u_n} \end{bmatrix}. \quad (0.7)$$

- Transposed Jacobian notation: $\frac{\partial f(u)}{\partial u} = J_f(u)^\top$.

Hessians

- For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **Hessian** captures second-order derivatives:

$$\nabla^2 f(u) = \begin{bmatrix} \frac{\partial^2 f}{\partial u_1^2} & \dots & \frac{\partial^2 f}{\partial u_1 \partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial u_n \partial u_1} & \dots & \frac{\partial^2 f}{\partial u_n^2} \end{bmatrix}. \quad (0.8)$$

- Symmetry: If second derivatives are continuous, $\frac{\partial^2 f}{\partial u_i \partial u_j} = \frac{\partial^2 f}{\partial u_j \partial u_i}$ (Schwarz's theorem).
- Hessian as Jacobian: $\nabla^2 f(u) = J_{\nabla f}(u)$.

Differentiability

- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **differentiable** at u if

$$\lim_{v \rightarrow u} \frac{\|f(u) - f(v) - A(u - v)\|}{\|u - v\|} = 0,$$

where $A = J_f(u)$ (the Jacobian).

- For scalar f , the derivative is $\nabla f(u)^\top$.
- Continuously differentiable** functions have continuous partial derivatives.

0.1.3 The Multivariable Chain Rule

Chain Rule for Compositions

- For $f = g \circ h$, where $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $g : \mathbb{R}^k \rightarrow \mathbb{R}$, we have

$$\nabla f(u) = J_h(u)^\top \nabla g(h(u)).$$

- For vector-valued $g : \mathbb{R}^k \rightarrow \mathbb{R}^m$,

$$J_f(u) = J_g(h(u))J_h(u). \quad (0.9)$$

Matrix Derivative of an Affine Transformation

- For $z = Wu + b$ and $y = g(z)$, the derivative with respect to W is

$$\frac{\partial y}{\partial W} = \frac{\partial y}{\partial z} u^\top. \quad (0.10)$$

Jacobian Vector Products (JVP) and Vector Jacobian Products (VJP)

- For composite $f = h_L \circ \dots \circ h_1$, let

$$g_\ell(u) = h_\ell(h_{\ell-1}(\dots(h_1(u))\dots)).$$

for each $\ell = 1, \dots, L$. Then, by recursive application of (0.9), we obtain

$$J_f(u) = J_{h_L}(g_{L-1}(u)) J_{h_{L-1}}(g_{L-2}(u)) \dots J_{h_1}(u). \quad (0.11)$$

Note that from the definition of the Jacobian, the j -th column of $J_f(u)$ is the m dimensional vector

$$\frac{\partial f(u)}{\partial u_j} = \left(\frac{\partial f_1(u)}{\partial u_j}, \dots, \frac{\partial f_m(u)}{\partial u_j} \right) = J_f(u) e_j,$$

where e_j is the j -th unit vector of appropriate dimension. Therefore, the **JVP** $\partial f(u)/\partial u_j$ can be computed recursively via

$$w_\ell = J_{h_\ell}(g_{\ell-1}(u)) w_{\ell-1},$$

starting with $w_0 = e_j$ and $g_0(u) = u$.

- On the other hand, since the i -th row of $J_f(u)$ is the gradient $\nabla f_i(u)$, we have

$$\begin{aligned} \nabla f_i(u) &= e_i^\top J_f(u) \\ &= \left[\dots \left[\left[e_i^\top J_{h_L}(g_{L-1}(u)) \right] J_{h_{L-1}}(g_{L-2}(u)) \right] \dots \right] J_{h_1}(u). \end{aligned} \quad (0.12)$$

That is, for each $i = 1, \dots, m$, the **VJP** $\nabla f_i(u)$ can be obtained recursively via

$$w_\ell^\top = w_{\ell-1}^\top J_{h_{L-\ell+1}}(g_{L-\ell}(u)),$$

starting with $w_0 = e_i$ and $g_0(u) = u$.

0.1.4 Taylor's Theorem

Univariate Case

- A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **k -times continuously differentiable** on an open interval $\mathcal{U}$ if all derivatives up to order k exist and are continuous on $\mathcal{U}$.

- **Taylor's Theorem in $\mathbb{R}$:** For k -times continuously differentiable univariate real-valued function f and $u, v \in \mathcal{U}$, we have

$$\begin{aligned} f(u) &= \sum_{i=0}^k \frac{(u-v)^i}{i!} \frac{d^i f(v)}{du^i} + O(|u-v|^{k+1}) \\ &= P_k(u) + O(|u-v|^{k+1}). \end{aligned} \quad (0.13)$$

where the **Taylor Polynomial**

$$P_k(u) = \sum_{i=0}^k \frac{(u-v)^i}{i!} \frac{d^i f(v)}{du^i},$$

and the Big-O notation $O(r^k)$ represents a function that, as $r \rightarrow 0$, satisfies $|O(r^k)| \leq C|r^k|$ for some constant $C > 0$, indicating that the remainder term, $f(u) - P_k(u)$, vanishes at least as fast as r^k .

- **Linear approximation ($k = 1$):** $f(u) \approx P_1(u) = f(v) + (u-v)f'(v)$.
- **Quadratic approximation ($k = 2$):** $f(u) \approx P_2(u) = f(v) + (u-v)f'(v) + \frac{(u-v)^2}{2}f''(v)$.

Multivariate Case

- **Multi-index Notation:** For $\alpha = (\alpha_1, \dots, \alpha_n)$:

$$|\alpha| = \sum_{i=1}^n \alpha_i, \quad \alpha! = \prod_{i=1}^n \alpha_i!, \quad u^\alpha = \prod_{i=1}^n u_i^{\alpha_i}.$$

- Higher-order partial derivative:

$$D^\alpha f(u) = \frac{\partial^{|\alpha|} f(u)}{\partial u_1^{\alpha_1} \cdots \partial u_n^{\alpha_n}}.$$

- **Taylor's Theorem in $\mathbb{R}^n$:** For k -times continuously differentiable multivariate real-valued function f and $u, v \in \mathcal{U}$:

$$f(u) = \sum_{\alpha: |\alpha| \leq k} D^\alpha f(v) \frac{(u-v)^\alpha}{\alpha!} + O(\|u-v\|^{k+1}). \quad (0.14)$$

- **Key Approximations:**

- Linear (first-order) approximation around v :

$$f(u) \approx P_1(u) = f(v) + (u-v)^\top \nabla f(v).$$

- Quadratic (second-order) approximation around v :

$$f(u) \approx P_2(u) = f(v) + (u-v)^\top \nabla f(v) + \frac{1}{2}(u-v)^\top \nabla^2 f(v)(u-v).$$

Linear Approximation with Jacobians and Hessians

- For differentiable $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with Jacobian J_f , an approximation of $f(u)$ around v is

$$\tilde{f}(u) = f(v) + J_f(v)(u - v). \quad (0.15)$$

- For twice differentiable $g : \mathbb{R}^n \rightarrow \mathbb{R}$ with Hessian $\nabla^2 g$, an approximation of the gradient $\nabla g(u)$ around v is

$$\tilde{\nabla} g(u) = \nabla g(v) + \nabla^2 g(v)(u - v). \quad (0.16)$$

Exercise 0.1.4.1

1. Find the Taylor series expansion of the function $f(x) = e^{2x}$ centered at $x = 0$ (Maclaurin series) up to the quadratic term (x^2).
2. Use your result to approximate the value of $f(0.2)$ and compare with the true value $e^{2 \times 0.2}$.